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Towards reasoning-empowered task-oriented communication for agent networks

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Agentic AI is fundamentally transforming wireless communication by reshaping static architectures into an autonomous, cognitive fabrics, ultimately paving the way for *agent networks*: distributed ecosystems of intelligent agents capable of perception, reasoning, and collaborative action across physical and digital domains. In such environments, communication evolves beyond transmitting bits to conveying intent, context, and task-relevant semantics that drive collective intelligence. As the key enabler of agent networks, *Task-Oriented Communication (TOC)* aligns transmission objectives with task execution, ensuring that agents are capable of sharing information most relevant for joint reasoning and decision-making. This paper examines the development of TOC, tracing its evolution from discriminative and generative paradigms while highlighting their inherent reactivity and limited support for autonomous agent behaviors. To address these limitations, we envision a reasoning-empowered TOC framework, where communication becomes a reasoning-driven process, enabling agents to autonomously determine not only *what*, but also *when* and *why* to communicate. This paper concludes by identifying critical challenges and outlining future research pathways towards enabling scalable, intelligent agent networks through reasoning-empowered TOC systems.

The fusion of ubiquitous wireless connectivity and advanced intelligence is set to redefine the cyber-physical landscape, heralding an era of AI-driven networks[1,2], which are expected to evolve to autonomous AI agents[3]. These agents, capable of perceiving based on multi-modal data from diverse environments, reasoning and planning for sophisticated tasks, will seamlessly bridge digital and physical realms[4,5]. This evolution marks a symbiotic convergence between wireless technologies and agentic AI, where communication transcends its traditional role as a data conduit to become the cognitive fabric of intelligent autonomy[6]. In such dynamic and hyper-connected environments, communication evolves from merely linking devices to empowering autonomous agents capable of reasoning, adaptation, and collaborative problem-solving.

The agent-centric paradigm is set to drive exponential growth in interconnected devices and intelligent services, and the resulting data deluge will strain existing wireless communication systems. Conventional wireless systems, rooted in Shannon's information theory and optimized for lossless transmission, face inherent conflicts between the stringent low-latency demands of real-time applications and the finite capacity constraints of traditional frameworks[7]. On the other hand, with the growing scarcity of spectrum resources, the pursuit of incremental gains in traditional task-agnostic metrics has become increasingly impractical and unsustainable[8]. Fortunately, intelligent services exhibit tolerance to graceful degradation, prioritizing task-relevant information over complete data fidelity, a characteristic that motivates a novel communication paradigm transcending bit-level reconstruction to focus on task-level effectiveness[9]. Consequently, task-oriented communication (TOC)[10] emerges as a transformative solution: by optimizing the transmission of only task-relevant information, it alleviates bandwidth constraints, reduces latency, and enhances efficiency for intelligent applications and services in future agent networks.

The evolution of TOC is closely intertwined with the progress of AI, advancing from *discriminative* models that transmit compact task-relevant features, to *generative* paradigms that convey latent representations, and more recently toward the emerging frontier of *agentic AI*, which embeds reasoning and autonomy directly into the communication process. In this paper, we examine this progression through the lens of AI evolution, classifying existing TOC frameworks into discriminative and generative paradigms and examining key methodologies, challenges, and existing advances.

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Table 1 | Existing literature, gaps, and how our reasoning-empowered TOC vision addresses them

Research Category	Primary methodology	Existing limitations & gaps	How Our vision addresses this
Feature-based TOC (e.g.,[9,10,68])	Information bottleneck; discriminative semantic encoders; deep JSCC.	Mainly design encoder–decoder mappings for fixed downstream tasks and treat communication as a reactive transmission stage, without closing the loop across perception, communication, and reasoning.	We embed these feature-based schemes into a **proactive cognitive loop**, where agents interpret intent, predict task evolution, and reason about what and when to communicate rather than always transmitting.
Generative model-based TOC (e.g.,[69–71])	Diffusion models; autoregressive models; latent-variable generative codecs.	Pursue high-fidelity reconstruction of rich content, but do not determine whether communication is necessary or integrate generative reconstruction with world-model prediction and reasoning-driven control.	Our framework uses generative models as tools inside a **world-model–driven reasoning loop**, forecasting outcomes, validating semantic consistency, and selectively communicating only the prompts or latent semantics that are truly required.
LLM/Agentic AI for Communication (e.g.,[20,72])	LLM-based agents; chain-of-thought reasoning; planning; world models.	Agentic LLM systems largely study perception and reasoning while assuming ideal communication, leaving the communication leg of the perception–reasoning–action loop unmodeled and detached from TOC and wireless constraints.	We provide the **first unified reasoning-empowered TOC architecture** that closes the perception–reasoning–communication loop by grounding LLM agents in goal-driven, TOC-aware communication policies under realistic PHY/MAC constraints.

Building upon these foundations, we present a forward-looking vision of *reasoning-empowered TOC*, a paradigm where communication becomes an active component of cognition, guiding *what*, *when*, and *why* information is exchanged. Table 1 outlines the contribution of reasoning-empowered TOC vision compared to the existing literature. Realizing this vision requires more than optimizing task-level efficiency; it calls for a holistic framework where autonomous agents can reason, interact, and evolve collectively. This necessity underpins the emergence of *Agent Networks*, representing the next evolutionary stage of intelligent connectivity that unites communication and cognition within a unified ecosystem.

Agent networks: vision and foundational technologies

The rapid fusion of agentic AI and wireless communication is motivating a shift from connected intelligence toward autonomous agents. In this forthcoming paradigm, *Agent Networks* represent a transformative vision where large populations of intelligent agents perceive, reason, and act collaboratively across physical and digital domains[2,3]. As illustrated in Fig. 1, agent networks are poised to play a pivotal role in future wireless systems, supporting goal-driven and task-oriented interactions that underpin emerging collective intelligence applications, such as smart cities, remote healthcare, and industrial automation. Among various enabling technologies, TOC stands out as a key enabler for agent networks by its in-depth integration and co-design in communication and intelligence, paving the way to data-driven low-latency intelligence autonomy.

From connected devices to autonomous agents

Future wireless systems will serve not merely as passive data pipelines but as cognitive ecosystems that host autonomous entities capable of making decisions, negotiating objectives, and adapting to changing environments. Each agent possesses a local perception of the world through multi-modal sensing and maintains a world model that enables reasoning and planning[11,12]. Through continuous semantic interactions, these agents form distributed intelligent organisms that cooperate and compete to accomplish complex missions, including holographic telepresence for remote collaboration, digital twin for urban Metaverses, and human-AI collaborative environments.

As agents interact with one another and their surroundings, their world models evolve dynamically, enabling continual learning and foresight. The network itself thereby transforms into an information-theoretic organism, where communication conveys intent, belief, and reasoning rather than mere bits[9,13]. This paradigm redefines fundamental design principles: communication becomes a vehicle for cognition, while cognition, in turn, dictates *what*, *when*, and *why* to communicate.

Foundational enabling technologies for agent networks

Achieving agentic autonomy requires breakthroughs across multiple technological fronts. Several emerging technologies are converging toward this vision:

- **Task-oriented communication**: Moving beyond bit-level transmission, TOC focuses on exchanging only information that is explicitly relevant to task objectives, such as control decisions, inference outcomes, or intent-aligned features. By abstracting communication around task utility rather than raw data fidelity, TOC enables agents to reduce bandwidth consumption, improve robustness, and coordinate more efficiently, thereby forming the informational backbone of goal-driven agent interactions[10].
- **Multi-modal perception**: Recent advances in foundation models allow agents to perceive, fuse, and reason across heterogeneous modalities including vision, audio, text, and sensor data. Such multi-modal perception grounds agent behavior in a richer understanding of the physical and semantic environment, supporting more accurate situation awareness, intent recognition, and contextual reasoning that are critical for downstream communication and decision-making[14].
- **Edge intelligence**: Edge computing infrastructures, together with federated and split learning paradigms, provide decentralized computational capabilities that enable agents to train, adapt, and collaborate close to data sources. This distributed intelligence reduces end-to-end latency, alleviates backhaul congestion, and enhances privacy preservation, while supporting scalable multi-agent systems that can operate under heterogeneous resource and network constraints[15,16].
- **Reasoning and planning with large-scale models**: Large language models (LLMs) and world-model-based planners equip agents with advanced reasoning, abstraction, and planning abilities, allowing them to translate high-level goals or natural-language intents into structured sequences of actions and communications. These capabilities enable agents to anticipate future states, evaluate alternative strategies, and proactively coordinate communication and control decisions rather than reacting myopically to instantaneous observations[17].

Roles of TOC in agent networks

Within this technological landscape, TOC emerges as a particularly promising foundation. By transmitting only the information that is directly relevant to achieving a specific task, TOC bridges data-centric connectivity with goal-centric intelligence[9,10]. In agent networks, TOC will provide the semantic substrate through which agents align their world models, exchange intent, and refine cooperative strategies without overwhelming the channel with irrelevant data[13,18].

Recent works have extended this idea toward semantic-driven AI agent communication, highlighting architectures and adaptation mechanisms for dynamic, resource-constrained environments[19,20]. Nonetheless, existing TOC schemes are tailored for static tasks and limited network contexts, while lacking the reasoning, adaptability, and autonomy demanded by agentic systems. These limitations motivate the development of reasoning-empowered TOC, where communication itself becomes a cognitive process tightly coupled with perception, planning, and decision-making. Such

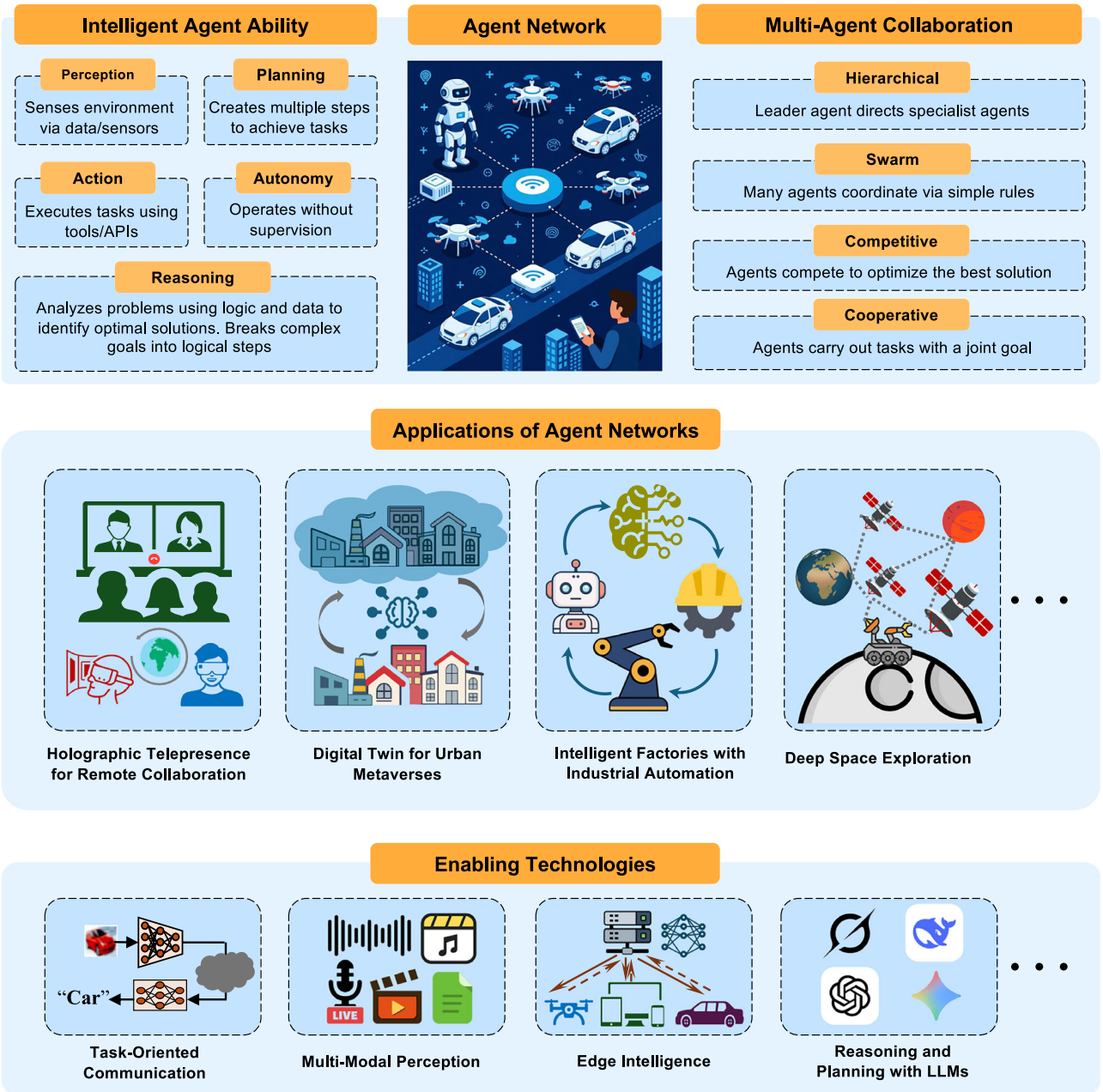


Fig. 1 | Overview of envisioned agent networks, illustrating the core capabilities of AI agents, characteristic multi-agent collaboration paradigms, transformative application scenarios, and pivotal technologies, with task-oriented communications as the key enabler for intelligent wireless ecosystems.

evolution will ultimately enable fully autonomous, self-organizing agent networks.

Existing task-oriented communication

From data-oriented to task-oriented communication

Conventional wireless systems were built upon the *data-oriented paradigm*, rooted in Shannon's information theory, where the objective was to transmit raw data with maximal fidelity, optimizing metrics such as throughput, bit error rate, and spectral efficiency[21]. This design philosophy successfully supported decades of digital communication evolution, from early telephony to broadband multimedia transmission, under the assumption that accurate data delivery inherently ensures application performance.

However, as wireless networks evolve into the *AI-driven era*, this assumption no longer holds. Intelligent and autonomous systems needs to operate within environments characterized by massive data generation, distributed computation, and real-time decision requirements[2,9]. In such settings, the objective of communication is no longer the faithful reconstruction of data; instead, it should target for the efficient exchange of information that directly serves the task at hand. This transformation redefines communication as an integral part of the cognitive loop, where each transmission contributes to collective reasoning, perception, or control. Consequently, the traditional notion of *data-oriented communication* is being replaced by TOC, which prioritizes task-relevant semantics and measurable task performance. It has been further emphasized that the importance of interpretability and effectiveness in semantic communications, bridging the gap between human-understandable semantics and machine-level reasoning[22].

As shown in Fig. 2, rather than treating all bits as equally valuable, TOC prioritizes extracting and transmitting only task-relevant information, leveraging principles such as the information bottleneck (IB) theory[23] to

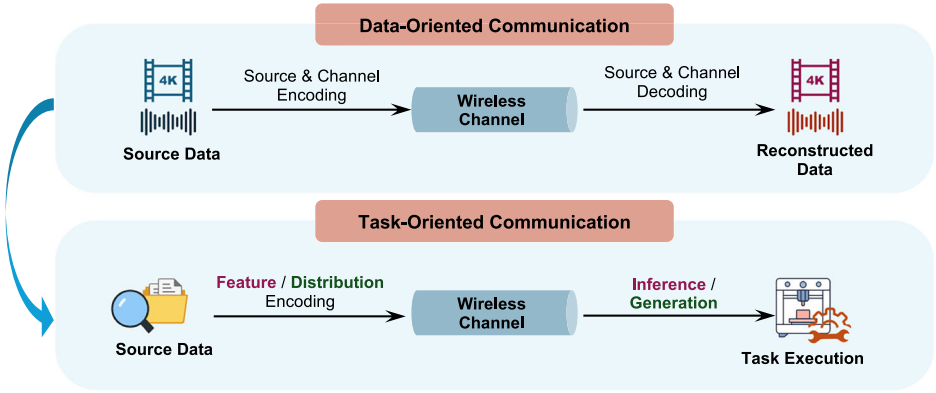


Fig. 2 | Illustration of the paradigm shift from data-oriented communication to task-oriented communication.

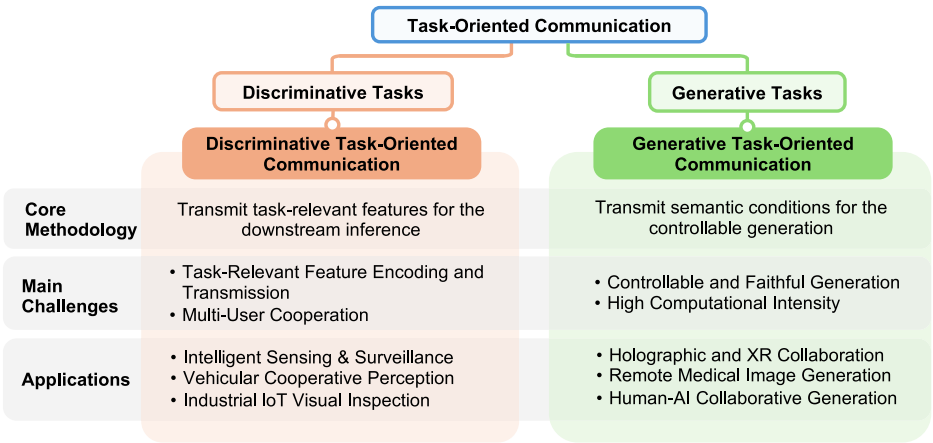


Fig. 3 | Comparison of discriminative and generative task-oriented communications.

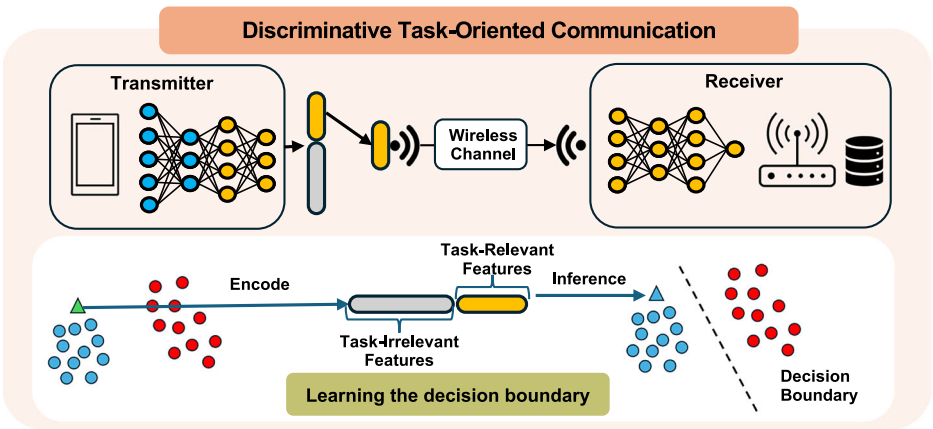


Fig. 4 | Illustrative example of discriminative task-oriented communications using classification as a discriminative task. The transmitter encodes data into task-relevant features, transmits them over an optimized wireless channel, and the receiver performs inference based on the received features and learned decision boundaries.

maximize the mutual information between the transmitted representation and the task objective while minimizing redundancy. As depicted in Fig. 3, existing TOC research can be broadly categorized into two complementary paradigms based on the task types: *discriminative TOC* for inference-driven tasks and *generative TOC* for generation-oriented tasks. The former emphasizes compact, task-relevant feature transmission for efficient downstream inference or decision-making, while the latter integrates generative models to convey high-level semantic representations that allow the receiver to reconstruct or synthesize task-relevant content[24]. Together, these approaches mark the transition from *bit-level precision* to *task-level effectiveness*, laying the groundwork for the reasoning-empowered TOC frameworks discussed in later sections[19,25].

Discriminative TOC: a feature-centric approach

Discriminative TOC marks a pivotal evolution in wireless systems, moving beyond traditional data transmission to extracting and conveying task-relevant features for efficient downstream inference. As depicted in Fig. 4, discriminative models are employed at the transmitter to distill high-dimensional source data (e.g., images or sensor readings) into compact and task-relevant feature vectors that directly support cooperative decision-making or inference at the receiver. This approach minimizes bandwidth usage by prioritizing features that maximize inference accuracy, while mitigating channel impairments through robust encoding techniques. The underlying technical methodology relies on the IB principle, which is realized by training DNNs as classifiers or regressors to learn an optimal low-

dimensional latent representation. This process forces the network to define a clear decision boundary in the feature space, ensuring the semantic content is highly discriminative. By leveraging discriminative AI, TOC paves the way for advancements in applications ranging from remote sensing to vehicular cooperative perception, where timely and accurate discriminative task completion is critical[10,26].

Despite their advantages, delivering real-time, high-quality AI services via discriminative TOC systems relies on addressing two principal challenges: extracting and transmitting task-relevant information across wireless channels, and cooperative inference with multiple users. Existing works have addressed these issues, such as IB methods for edge inference and maximal coding rate reduction techniques for multi-device feature sharing. Among the multiple challenges faced by discriminative TOC, we highlight two key aspects—feature encoding and cooperative inference, which will be discussed in the following subsections.

Task-relevant feature encoding and compression. Extracting task-relevant information involves identifying and isolating the relevant features necessary for the receiver to accomplish a specific task. Designing such encoders requires understanding the requirements of the task, modeling the source data statistical properties, and adapting to dynamic conditions (e.g., distribution shift), while avoiding excessive computational overhead that could strain resource-constrained devices.

TOC was initially introduced in ref. 10 for edge inference systems to encode the task-relevant feature by leveraging IB theory[23]. To derive informative yet compact features, IB maximizes the mutual information between the inference outcome and encoded features while minimizing the mutual information between input data and features. While the compression reduces communication overhead in edge inference, it also increases vulnerability to distortions caused by transmission unreliability or information obfuscation. Consequently, alongside feature informativeness, retaining redundancy is crucial to safeguard task-relevant information against potential distortions[27]. Robust IB addresses this by delineating features into task-relevant information, task-irrelevant information, and protective redundancy[28]. Moreover, single-user TOC systems in edge inference necessitate encodings invariant to input data distribution shifts, such as domain or semantic variations. Augmenting IB with invariant risk minimization bolsters generalization to domain shifts and detection of semantic shifts[29].

Task-relevant feature transmission. Beyond informativeness, robustness, and invariance in encoded features, TOC systems require that transmitted features are recognizable and utilizable by the receiver. To address this, end-to-end (E2E) architectures with joint optimization of on-device encoders and server-based neural networks have demonstrated efficacy in task performance and bandwidth conservation[24]. Unlike conventional communication systems that employ explicit arithmetic source and channel coding schemes, TOC systems often encode task-relevant features implicitly using neural networks with the E2E optimization. It introduces additional practical challenges in feature transmission, including the handling of floating-point tensors, fluctuations in channel conditions, and dynamic real-world network connections.

First, the features produced from neural encoders are typically floating-point tensors, which pose implementation challenges on the current radio frequency (RF) systems. Probabilistic vector quantization has been integrated into task-relevant feature encoding to incorporate digital modulation[28,30]. In parallel, a shared semantic-aware codebook was designed to improve representation consistency and coding efficiency across users[31].

Real-time deployment concerns, including incomplete coverage and server malfunctions, necessitate cross-model communication across inference systems from different providers. Aligning disparate feature spaces ensures that task-relevant information is identifiable and cross-model usable[32]. In dynamic wireless topologies, local training for each network connection is infeasible, and task specifics may be unavailable pre-connection. To address this challenge, a self-supervised mutual information maximization approach together with fine-tuning was proposed to train TOC for edge inference[33]. Besides, various adaptive techniques in learning-based joint source-channel coding (JSCC) schemes, such as attention mechanisms and hypernetwork, can also be incorporated in the existing single-user TOC systems to enhance their adaptability[34,35].

Task-oriented cooperative inference. While the previous single-user scenario is fundamental in many applications, the constrained computational resources and limited sensing capability of a single device make it inadequate for complex tasks with more complicated data sources. In the multi-user scenario, the correlation among the input data of different devices results in additional redundancy and latency in transmission. Although involving multiple edge devices provides richer sensed data and computational resources, it also amplifies communication overhead and latency with the number of involved users. Device cooperation is needed to mitigate this issue by exploiting correlations among sensed data, extracted features, and communication resources. Within the TOC framework, the goal is to efficiently extract sufficient task-relevant information in multi-device settings, formulated as a distributed feature coding problem.

This is achieved by extending IB theory to the distributed setting, which removes task-irrelevant correlated information across input views[26,36]. For camera-equipped edge devices, edge video analytics exemplifies multi-device edge AI, requiring cooperative inference from video frames to server-side decisions[37]. Task-oriented frameworks remove spatial and temporal video redundancy, transmitting minimal information to conserve bandwidth. However, system design inconsistencies arise between E2E task effectiveness and traditional communication modules. For instance, conventional multiple-input multiple-output (MIMO) precoding is designed to maximize throughput and reliability, prioritizing generic data transmission metrics over task-specific performance objectives. To resolve this, maximal coding rate reduction has been adopted to align feature encoding and precoding optimization toward maximizing inference accuracy[38,39].

Generative TOC: a distribution-centric paradigm

Unlike discriminative TOC, which optimizes transmitted features for downstream prediction, generative TOC integrates a generative model into the communication loop to reconstruct, complete, or synthesize task-relevant content from a received semantic condition. This is achieved by leveraging specific generative architectures: variational autoencoders (VAE)[40], and generative adversarial networks (GANs)[41] learn the underlying semantic manifold and data distribution; while diffusion models[42], auto-regressive models[43], and multimodal LLMs[44] achieve synthesis through iterative denoising. The transmitter no longer seeks to preserve raw pixels or modalities; instead, it conveys intent that, combined with the receiver's generative prior, produces outputs sufficient for the task. This critical ability to accurately model data distributions forms the conceptual basis for the World Model central to our proposed reasoning framework.

As shown in Fig. 5, the transmitter captures the essence of the source distribution and distills it into a streamlined condition, such as a prompt, multimodal embedding, or JSCC-learned latent. This compact form is transmitted across the channel, carrying just enough semantic depth to spark creation at another end. The receiver's generative model expands the received condition into vivid, task-aligned outputs or faithfully reconstructs relevant data, seamlessly supporting processes like object detection, planning, or control, leveraging its ability to intelligently fill transmission gaps. Although generative TOC offers strong expressive power, it also introduces new challenges in reliability and efficiency. The following subsections highlight two key aspects that remain central to its realization: controllable reconstruction and efficient generative transmission.

Reliable and controllable generation. Despite the promise of generative TOC, guaranteeing controllability and faithfulness remains a key challenge. The receiver must accurately reflect the transmitter's intent

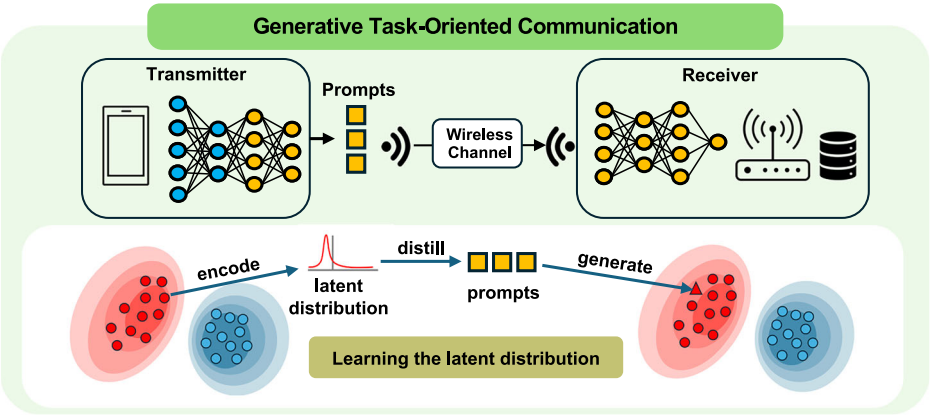


Fig. 5 | Illustrative example of generative task-oriented communications. The transmitter estimates and encodes a task-conditional latent distribution, sends it over a wireless channel, and the receiver performs semantic generation to produce task-relevant output.

Table 2 | Comparison of TOC paradigms and novelty of reasoning-empowered TOC

	Discriminative TOC	Generative TOC	Reasoning-Empowered TOC
Objective	Transmit compact task-relevant features for inference	Transmit semantic latent conditions for controllable generation	Agents autonomously determine *what*, *when*, and *why* to communicate through reasoning
Communication Trigger	Reactive: triggered by new data input	Reactive: triggered by new data input	**Proactive**: intent-driven, prediction-driven, and context-aware
Information Basis	Low-dimensional discriminative features	Latent distributions or prompts	Intent, belief state, world-model prediction, reasoning traces
Required Intelligence	Task-specific encoder/decoder networks	Generative models (diffusion, autoregressive, multimodal LLMs)	**Agentic AI**: intent parsing, autonomous problem formulation, foresight via world models
Level of Autonomy	None (task predefined)	Limited (reactive generation)	**High autonomy**: self-driven planning, adaptation, and communication decisions

even under channel corruption, avoid hallucinations, preserve safety-critical semantics, and align with task performance rather than perceptual fidelity. Generative TOC requires that the receiver's outputs remain both controllable and faithful to transmitted semantics, even under channel noise and adversarial perturbations[45]. Standard generators, typically trained with clean inputs, often drift or hallucinate when conditions are noisy or incomplete. This motivates channel-aware training and task-aligned evaluation metrics beyond pixel fidelity[46–48]. Complementarily, a robust image semantic communication system was introduced with a multi-scale vision transformer[49], which enhances resilience to channel corruption and cross-domain generalization.

Recent advances address this challenge along several directions. Researchers have aligned objectives with task metrics by incorporating semantic-preservation terms for safety-critical classes, while others adopted channel-aware conditioning by training with realistically corrupted inputs or denoising the condition prior to generation. At the same time, knowledge grounding has been introduced, where source, task, and channel knowledge are partitioned and exploited to constrain decoding, and lightweight validation at the receiver further improves robustness through rule checks, self-consistency tests, or hybrid signaling that couples hard constraints with soft prompts. Together, these strategies shift evaluation away from pixel-level fidelity and toward semantic- and task-aware metrics that more accurately capture meaning preservation and task success[47,48,50].

Resource-efficient generative transmission. Besides the challenges of controllability and faithfulness, the computational intensity of modern generators presents a major barrier for practical deployment. The goal is to reduce E2E task latency and energy consumption while maintaining task quality, rather than merely minimizing FLOPs[47,48].

Efforts to mitigate computational cost span three interconnected levels. At the model level, compression techniques such as distillation, pruning, quantization, and low-rank adaptation reduce complexity while preserving adaptability. At the sampling level, few-step solvers for diffusion models—including higher-order stochastic differential equation/ordinary differential equation (ODE/SDE) schemes, consistency objectives, and progressive distillation—collapse tens of iterations into only a handful with limited quality degradation[51–53]. Furthermore at the decoding stage, speculative or assistant decoding accelerates generation by drafting with a smaller model and verifying with a larger one, while early-exit and caching exploit intermediate confidence and computation reuse[54]. Together, these optimizations are complemented by system-level designs such as edge–cloud co-inference, opportunistic knowledge placement, and joint scheduling of rate and compute, which collectively ensure that semantic generation can meet latency and energy budgets without sacrificing task performance.

Although generative communication techniques enhance expressiveness and enable the reconstruction of task-relevant information, they remain largely reactive—responding to received inputs rather than proactively initiating or managing exchanges. To move beyond this limitation, future systems must embed reasoning capabilities directly into the communication process, enabling networks to determine not only *what* information to transmit, but also *when* and *why* it should be shared.

Reasoning-empowered task-oriented communication

While existing TOC frameworks—whether discriminative or generative—have greatly advanced communication efficiency, their operations remain fundamentally reactive[9]. They rely on predefined tasks and external triggers, lacking the capability to reason about goals, anticipate context, or adapt their communication strategies autonomously. As networks evolve toward intelligent, self-organizing systems, such passivity becomes a limiting factor.

To overcome this constraint, *reasoning-empowered TOC* introduces agentic intelligence into the communication loop, embedding sensing, planning, and reasoning directly within the transmission process[25]. This paradigm redefines the essence of communication—from optimizing the delivery of task-relevant information to determining *what*, *when*, and *why* to communicate in pursuit of self-formulated objectives. Table 2 compares

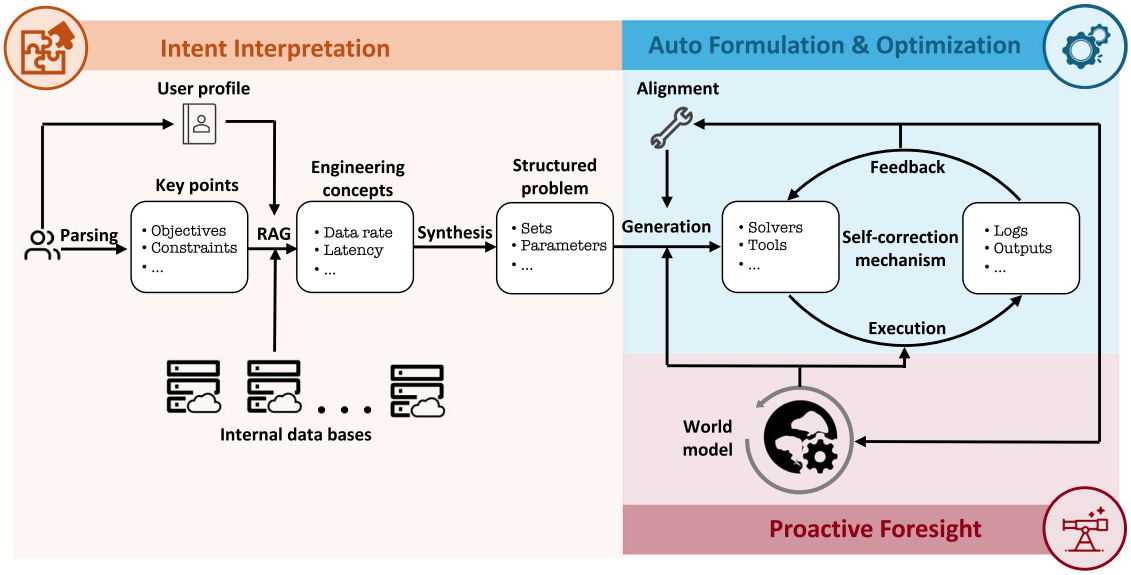


Fig. 6 | Adaptive pipeline of reasoning-empowered TOC.

existing discriminative and generative TOC with the envisioned reasoning-empowered TOC.

The agentic TOC paradigm is actualized through a continuous cognitive loop, whose methodology is illustrated in Fig. 6. This adaptive pipeline is designed to manage tasks that are initially ambiguous or undefined with the following looped stages:

- **Intent interpretation:** The agent processes unstructured or qualitative user intents to formulate a formal, machine-readable problem definition. This process involves parsing linguistic expressions, identifying objectives, constraints, and contextual cues, and mapping them to precise engineering concepts. By leveraging retrieval-augmented reasoning and domain knowledge bases, the agent ensures that the interpreted intent is both technically feasible and contextually grounded. This stage thus bridges human-level objectives and system-level design parameters, laying the foundation for autonomous problem formulation.
- **Automated formulation and optimization:** Upon receiving the structured problem definition, the agent autonomously formulates and optimizes a TOC solution tailored to the task requirements. It selects suitable architectural paradigms, resource allocation policies, or co-design strategies while jointly balancing efficiency, robustness, and scalability. A reasoning-based feedback loop allows the agent to assess performance, correct errors, and adapt its solution to changing conditions, turning communication design into a self-improving cognitive cycle.
- **Proactive foresight:** By predicting future environmental states and task outcomes, the agent leverages an internal world model to anticipate changes and proactively adapt its communication strategy. This predictive reasoning enables dynamic re-optimization before performance degradation occurs, ensuring stability under uncertainty such as channel variation or user mobility. The foresight stage continuously integrates feedback from real-world observations to update the world model, enhancing long-term autonomy and adaptability. In this way, the communication system evolves from reactive optimization to proactive, reasoning-driven decision-making.

From natural language to structured objectives

Intent interpretation is the foundational stage of the loop, where the agent converts an unstructured, qualitative, user intent into a formal, machine-readable problem definition, i.e., the task recognition. Unlike existing schemes that operate on predefined tasks, this stage reasons about ambiguous inputs to define the underlying technical objective itself. For example, a high-level intent such as "stable video conferencing" for a mobile user, traditionally mapped to a static QoS profile, is interpreted by the agent into a specific, dynamic TOC design problem[55].

The interpretation process is executed by an agent, typically deployed at the base station (BS). It begins by parsing the user's intent into its key components—objectives, constraints, and qualitative descriptors, and mapping these linguistic elements onto precise engineering concepts. To ensure this mapping is both contextually relevant and factually grounded, the agent employs Retrieval-Augmented Generation (RAG) to dynamically source information from external knowledge bases (e.g., 3GPP technical specifications) and internal user profiles[55,56]. Subsequently, the agent synthesizes these inputs to identify candidate technical solutions (e.g., adaptive beamforming, semantic source coding) and formulate a trade-off space (e.g., video resolution vs. power consumption). By weighing these factors, the agent constructs the final output: a structured problem definition, complete with optimization objectives, variables, and constraints. This deterministic output serves as the direct input for the subsequent stage of execution, bridging the gap between ambiguous human intent and precise engineering formulation.

From formulation to solution: agent-driven execution

Upon receiving the formal problem definition, the agent proceeds to the Automated Formulation and Optimization stage. The function of this stage is to translate the defined objective into a concrete, optimized, and deployable TOC solution. Harnessing its reasoning capabilities, the agent formulates a solvable problem that can span a spectrum from high-level architectural design to fine-grained resource management. For instance, depending on the requirements of the task, the agent may formulate an architectural design problem, selecting a foundational communication paradigm and defining its core components. Alternatively, it might formulate a resource optimization problem, often modeled as a constrained optimization task (e.g., a mixed-integer program), to allocate network resources like power and bandwidth under constraints derived from the user's intent. In complex cases, the agent can formulate a co-design problem, jointly optimizing architectural parameters and resource allocation policies to achieve a global optimum. This automated process enables continuous self-optimization in response to dynamic conditions, such as changes in network topology or task priorities.

Given the structured output, the agent shall transform it into a "five-element" formulation[57]. Based on this formulation, the agent proceeds to generate solver code and invoke a tool for execution. To mitigate plausible-yet-incorrect hallucinations in the code, alignment techniques like KTO

(Kahneman-Tversky Optimization) can be employed to enhance the robustness. Furthermore, the system automatically analyzes execution logs and results, triggering a self-correction mechanism upon detecting any errors or illogical outputs. The agent then consolidates the original problem, its own generated formulation, the code, and the error logs as a new input for self-reflection, deciding whether to revise the modeling layer (the formulation) or the coding layer (the code) before re-attempting. This forms a complete feedback and correction loop[57,58].

Proactive planning and adaptation with world models

The culminating stage of the cognitive loop is Proactive Foresight, which is enabled by an internal world model. This predictive model allows the agent to simulate future states and task outcomes, transitioning it from a reactive problem-solver to a proactive strategist. By forecasting environmental dynamics, such as channel state information (CSI) variations or user mobility patterns, the agent can preemptively design or reconfigure TOC solutions to be robust against anticipated changes. This foresight is sustained by a continuous learning cycle: the world model is refined with feedback from real-world observations, leading to progressively more accurate predictions and improved long-term performance. This predictive capability is also the key enabler of the framework's inner adaptation loop, where forecasted environmental shifts trigger a direct, rapid re-engagement of the Automated Formulation and Optimization stage without requiring a new high-level intent[12,59].

To implement the world model, we propose a self-supervised approach based on the Joint-Embedding Predictive Architecture (JEPA)[12]. Unlike generative methods that focus on pixel-level reconstruction, JEPA operates by predicting future representations in an abstract latent space. This objective incentivizes the model to learn the underlying dynamics of the environment in an information-efficient manner. To address the inevitable mismatch between the pre-trained model and the live operational environment, parameter-efficient fine-tuning (PEFT) is employed for continuous online adaptation, using the discrepancy between predictions and observations as a feedback signal[60,61]. Finally, to ensure reliable decision-making under the inherent stochasticity of wireless environments, the agent's policy optimization incorporates Robust Reinforcement Learning (RRL). By training the policy against a virtual adversary that represents model uncertainty, RRL produces conservative yet effective strategies with performance guarantees, ensuring system resilience even when predictions are imperfect.

Interaction between reasoning and physical layer

The performance of reasoning-empowered TOC hinges on a profound interaction with the physical layer. This integration transcends traditional protocol layering by creating a continuous, closed-loop dependency. In this framework, the reasoning system dynamically manages the communication channel, while the channel, in turn, supplies vital real-time context to the world models. This synergistic relationship enhances adaptability and efficiency, ensuring effective performance in evolving wireless environments.

Reasoning framework as a physical layer tool master. In the proposed framework, the reasoning agent acts as a powerful tool master, strategically utilizing advanced physical layer techniques to achieve high-level semantic objectives. As presented in the Agent-driven Execution subsection, the agent translates complex intent into precise, executable physical layer configurations, optimizing performance across the entire system rather than just maximizing raw data rate. For capacity and interference management, the agent can actively leverage massive MIMO[62] and multi-band networks to secure the necessary capacity for distributing large state information and world model updates. Additionally, it can utilize rate-splitting multiple access (RSMA)[63] to manage semantic interference by strategically separating common beliefs from private intent messages. Moreover, the reasoning framework can treat the channel itself as an optimizable element, using advanced control surfaces such as fluid antennas to actively and dynamically shape the wireless environment and maximize semantic signal strength directed toward the target agent.

Physical impairments as constraints and inputs. In practice, dynamic wireless channel conditions impose inherent constraints on the reliability of reasoning-empowered TOC. These environmental fluctuations not only challenge performance but also serve as a critical data source essential for effective uncertainty modeling and the robust learning of the world model. In particular, physical layer impairments, such as fading, ambient noise, and severe interference, introduce significant uncertainty that can severely degrade system performance. For instance, multipath fading causes inter-symbol interference by generating time-delayed signal replicas that interfere with the current symbol. When the communication channel distorts transmitted information, the agentic reasoning framework is compelled to operate on compromised data, leading to misinterpretations and potentially suboptimal decision-making. Consequently, the inherent unreliability of the channel presents a critical constraint for reasoning-empowered TOC, necessitating advanced strategies for mitigating these adverse effects.

Conversely, the physical channel also functions as a continuous source of valuable real-time input for the reasoning framework. Inputs such as channel state information (CSI) and signal-to-noise ratio (SNR) are continuously fed into the world model. This feedback loop enhances the model's ability to adapt quickly and accurately in real time. By continuously updating its understanding of current and anticipated wireless dynamics—such as user mobility patterns, traffic loads, and environmental conditions—the world model significantly improves its predictive capabilities.

Sustaining this channel-aware context enables the reasoning engine to proactively identify potential communication challenges, such as increased latency or reduced signal quality. This proactive approach empowers the planning module to select the most effective transmission policy in advance, ensuring that the system is prepared to respond swiftly to changing conditions before impairments occur. Consequently, this comprehensive framework can not only optimize overall communication efficacy but also ensures a more reliable and resilient networking experience in ever-changing environments.

Challenges and opportunities

Reasoning-empowered TOC transforms wireless networks from reactive frameworks into proactive cognitive systems. In this paradigm, communication itself becomes a reasoning action: each transmission reflects an agent's intent, belief, and decision. Such a shift enables collective intelligence but also introduces unique theoretical and practical difficulties that differ fundamentally from conventional bit-level communication and existing TOC.

Lack of theoretical foundations

Reasoning-empowered TOC with agentic AI diverges fundamentally from Shannon's theory by incorporating semantics, reasoning, and autonomy. Yet, it currently lacks a unified theoretical framework to quantify performance limits, such as capacity bounds that explicitly incorporate intent dynamics and closed-loop cognitive interactions. While recent studies on semantic communication provide initial insights into meaning-based metrics, extending these to agentic processes remains significantly underdeveloped[19,25,64]. Current frameworks struggle to quantify the informational utility for long-horizon planning or the stability of feedback loops between reasoning and transmission. Moreover, multi-agent coordination problems persist as open challenges, with ongoing research highlighting unresolved issues in emergent behaviors, scalability, and security within distributed reasoning systems where agent policies are non-stationary. These deep theoretical gaps call for innovative frameworks to establish unified mathematical foundations and fully harness the agentic AI potential in TOCs.

Stability of the reasoning-communication loop

A fundamental challenge lies in the stability of the reasoning-communication loop, particularly in deciding when and why communication should occur. Unlike conventional systems that transmit data deterministically whenever new observations arrive, reasoning-empowered agents must autonomously evaluate the value of information, determining whether an exchange genuinely improves collective understanding or decision quality. Triggering communication based on intent requires agents to estimate the potential impact of their messages on the reasoning processes of peers. However, because agents possess distinct, partially observable world models, these estimations are fraught with uncertainty. In decentralized settings, this can trigger semantic feedback loops, causing oscillatory over-communication or "overthinking" behaviors where agents continuously exchange refinements without converging on an action. Consequently, ensuring the stability and convergence of these asynchronous reasoning loops remains a critical open problem that demands novel control-theoretic approaches tailored for cognitive dynamics.

Multi-agent coordination under limited spectrum

In reasoning-empowered TOC, communication among agents is no longer a simple data exchange but a deliberation process about beliefs, plans, and goals. When many reasoning agents share the same spectrum, determining who should speak, what should be transmitted, and how to reconcile conflicting messages becomes extremely complex. The coordination mechanism must prioritize transmissions that contribute the most to collective reasoning to avoid semantic congestion caused by redundant intent updates. Along this line, it has been demonstrated that semantic-aware spectrum sharing in Internet of Vehicles via deep reinforcement learning[65], showcasing how learning-based coordination can enhance spectrum efficiency in multi-agent semantic environments. Designing efficient and fair spectrum allocation mechanisms that optimize for semantic value while managing resource constraints is a key challenge for reasoning-empowered TOC.

Trustworthiness and operational risks of agentic reasoning

The integration of LLM-based reasoning into TOC presents significant challenges concerning trustworthiness and operational integrity, particularly in light of the absence of established industry standards. As these advanced agentic models dictate communication policies in autonomous systems, their reliability becomes paramount. A primary risk arises from issues such as semantic ambiguity and hallucination, where the intents derived from LLMs may misalign with high-level safety constraints or stem from inaccurate contextual data. Such discrepancies can lead to miscommunication, raising ethical concerns and potentially jeopardizing safety.

To effectively mitigate these risks, it is essential to develop robust validation layers that implement stringent, predefined safety checks on the reasoning output before any physical actions are executed. These checks serve as gatekeepers, ensuring that all decisions made by the system adhere to safety protocols, thus minimizing the potential for adverse outcomes in high-stakes environments.

Furthermore, the reliance on black-box AI methods necessitates a focused approach to interpretability and security. Operators require transparent, traceable justifications for the communication decisions generated by these models to facilitate auditing and accountability. Additionally, implementing strong security protocols is crucial to defend against adversarial attacks aimed at manipulating the reasoning engine. Such attacks could exploit vulnerabilities, leading to the generation of intentionally misleading or misaligned intents that could undermine the system's integrity.

Feasibility and latency of LLM-based reasoning

The integration of complex, high-load models such as LLMs and world models demands a well-defined architectural strategy to ensure system feasibility while effectively addressing the inherent trade-offs between computational overhead and efficiency gains. A key approach to mitigating feasibility concerns lies in deploying these computationally intensive AI components exclusively on the network side, rather than on resource-constrained end devices. This architectural decision allows for more robust processing capabilities. Meanwhile, the effect of computational load on end-to-end latency must be carefully considered.

For real-time applications, the round-trip delay associated with agent data transmission, reasoning process execution, and communication policy optimization must not undermine the efficiency gains achieved through reasoning. Therefore, a critical area for future research involves developing hierarchical and partitioned architectures. This approach entails strategically distributing intelligence within the system—leveraging network-side LLMs for complex, long-horizon planning tasks while deploying lightweight, specialized neural modules directly on the agents. These specialized modules are designed for ultra-low-latency, reactive adjustments, ensuring prompt responses to dynamic environmental changes.

Furthermore, new unified performance metrics are essential to accurately quantify the trade-offs between semantic gains achieved through reasoning and the total end-to-end latency imposed by the network-side reasoning process. These metrics should encompass not only latency but also factors such as energy consumption and computational load, thereby providing a comprehensive evaluation framework.

System integration, standardization, and benchmarking

The existing layered communication structures and protocol stacks were primarily developed for data exchange at the bit level, focusing on efficient transmission without the need for higher-level reasoning capabilities. However, the integration of reasoning-empowered mechanisms and varying levels of task effectiveness fundamentally challenges these established design principles, necessitating a reevaluation of network architectures or potentially disruptive modifications[66,67]. This incompatibility arises largely because conventional systems were designed for modular, task-agnostic operations that operate independently of one another. In contrast, agentic TOC requires a comprehensive redesign that embeds AI-driven autonomy seamlessly across all layers—physical, data link, network, and application. Such extensive changes may not only disrupt backward compatibility but also considerably increase deployment complexity, requiring careful planning and execution to ensure effective integration.

Moreover, a significant gap exists in standardized protocols for the integration and validation of LLM-driven decision-making within communication systems. Currently, no comprehensive standards exist to certify LLM behavior for mission-critical wireless tasks, leaving a critical void in ensuring reliability and safety. As a consequence, future efforts must prioritize the establishment of certified reliability metrics and benchmarking protocols that can guide the deployment of these advanced systems.

This endeavor necessitates a shift away from traditional communication metrics toward the development of unified performance measures that effectively quantify not just the communication speed but also the AI's semantic success rate, decision latency, and energy efficiency within complex, dynamic, and multi-agent simulation environments.

Conclusion

In the culmination of wireless evolution, agent networks are envisioned the ultimate architecture for intelligent systems, where interconnected AI agents autonomously collaborate across distributed environments to achieve collective goals. These networks transcend traditional point-to-point communications, forming dynamic ecosystems that integrate perception, reasoning, and action in real-time.

Yet, realizing agent networks introduces profound challenges, including interoperability among heterogeneous agents, scalability in massive deployments, and security against adversarial manipulations in decentralized exchanges. TOC offers a promising pathway to mitigate these issues by prioritizing semantic efficiency—transmitting only task-relevant information to reduce overhead, enhance latency, and ensure robust coordination. Through reasoning-empowered TOC, agents can proactively infer needs, optimize transmissions, and predict

disruptions, transforming potential vulnerabilities into strengths for sustainable network performance.

In this paper, we have charted TOC's progression from discriminative and generative AI paradigms to agentic innovations. By proposing a cognitive loop methodology for reasoning-empowered schemes, we bridge TOC schemes with future agent networks. Finally, we explored the key challenges and open research directions to unlock transformative wireless intelligence and guide ongoing advancements in autonomous ecosystems.

Data availability

No datasets were generated or analysed during the current study.

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Competing interests

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